Summer Bootcamp – Session 2
Corrections to the Calculus and ODE Exercises

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Correction Guidelines

The solutions follow the numbering of the exercise sheet. Numerical values are rounded only at the end. For programming questions, the displayed formulas are the reference implementation; a vectorized implementation is equally valid.

1 Python and Elementary Financial Calculations

Exercise 1 – Coupon-bond yield, duration, and repricing. Differentiation gives

$$B'(y) = -4e^{-y} - 8e^{-2y} - 312e^{-3y}, \qquad B''(y) = 4e^{-y} + 16e^{-2y} + 936e^{-3y}.$$

Since $B'(y) < 0$, the equation $B(y) = 96.50$ has at most one solution. Moreover, $B(0) = 112$ and $B(0.2) \simeq 63.03$, so $[0, 0.2]$ brackets it. Bisection or Newton's iteration

$$y_{n+1} = y_n - \frac{B(y_n) - 96.50}{B'(y_n)}$$

gives

Result

$y = 0.0515698674$, i.e. a continuously compounded yield of 5.15699%. At this yield,

$$D_{\mathrm{mod}} = 2.88387666, \qquad \mathcal{C} = 8.49811801.$$

For a yield move Δy, duration and duration–convexity give

$$B_D = B(1 - D_{\mathrm{mod}}\Delta y), \qquad B_{DC} = B\left(1 - D_{\mathrm{mod}}\Delta y + \tfrac{1}{2}\mathcal{C}(\Delta y)^2\right).$$

Δy	Exact price	Duration	Duration–convexity
−100 bp	99.3244	99.2829	99.3239
−25 bp	97.1983	97.1957	97.1983
+25 bp	95.8068	95.8043	95.8068
+100 bp	93.7577	93.7171	93.7581

The quadratic correction is visibly more accurate, especially for 100 bp.

2 Explicit Derivatives

Exercise 2 – One-dimensional derivatives. The derivatives and domains are

$$f_1' = 6x\cos(3x^2 - 1), \quad \mathrm{Dom}(f_1) = \mathbb{R},$$

$$f_2' = e^{-3x}(2x - 3x^2), \qquad f_2'' = e^{-3x}(2 - 12x + 9x^2), \quad \mathrm{Dom}(f_2) = \mathbb{R},$$
$$f_3' = \frac{(1+x^2)/x - 2x\log x}{(1+x^2)^2}, \quad \mathrm{Dom}(f_3) = (0,\infty),$$
$$f_4' = \frac{e^x}{1+e^x}, \qquad f_4'' = \frac{e^x}{(1+e^x)^2}, \quad \mathrm{Dom}(f_4) = \mathbb{R},$$

Result

$f_5' = \dfrac{x}{\sqrt{1+x^2}}$ and $\mathrm{Dom}(f_5) = \mathbb{R}$.

Exercise 3 – Gradient, Hessian, and Jacobian. One obtains

$$\nabla f = \begin{pmatrix} 4x - y + 4 \\ -x + 6y \end{pmatrix}, \qquad H_f = \begin{pmatrix} 4 & -1 \\ -1 & 6 \end{pmatrix}.$$

Furthermore,

$$J_g(x,y) = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ 1 & -2y \end{pmatrix}.$$

Exercise 4 – Multivariate chain rule by two methods. Here

$$F(x,y) = (x+y^2)e^{xe^y}.$$

Direct differentiation yields

$$\partial_x F = e^{xe^y}\big[1 + (x+y^2)e^y\big],$$
$$\partial_y F = e^{xe^y}\big[2y + x(x+y^2)e^y\big].$$

On the other hand,

$$J_\Phi = \begin{pmatrix} 1 & 2y \\ e^y & xe^y \end{pmatrix}, \qquad \nabla q(u,v) = \begin{pmatrix} e^v \\ ue^v \end{pmatrix}.$$

Thus $\nabla F = J_\Phi^\top \nabla q(\Phi)$ reproduces the two formulas above. Equivalently,

$$D\Phi h = \begin{pmatrix} h_1 + 2yh_2 \\ e^y h_1 + xe^y h_2 \end{pmatrix}, \qquad Dq(u,v)k = e^v k_1 + ue^v k_2,$$

and substituting $(u,v) = \Phi(x,y)$ gives the same coefficients of h_1, h_2.

3 Explicit Integration

Exercise 5 – One-dimensional integrals. Direct integration, substitution, integration by parts, and a trigonometric substitution respectively give

Result

$$I_1 = 1, \qquad I_2 = \begin{cases} \dfrac{(1+T)^{1-r} - 1}{1-r}, & r \neq 1, \\ \log(1+T), & r = 1, \end{cases} \qquad I_3 = \frac{e-1}{2},$$
$$I_4 = -\frac{1}{4}, \qquad I_5 = \frac{\pi}{4}.$$

For I_2, the limit as $r \to 1$ follows from $\lim_{a\to 0}((1+T)^a - 1)/a = \log(1+T)$. For I_4, the boundary term $x^2 \log x/2$ tends to zero at the origin.

Exercise 6 – The Gaussian integral. Tonelli's theorem allows the nonnegative integrand to be written as

$$G^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,\mathrm{d}x\,\mathrm{d}y.$$

With $x = r\cos\theta$, $y = r\sin\theta$ and Jacobian r,

$$G^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi\left[-\frac{1}{2}e^{-r^2}\right]_0^{\infty} = \pi.$$

Since $G > 0$,

Result

$G = \sqrt{\pi}$, and, after $x = \sqrt{2}\sigma u$,

$$\int_{-\infty}^{\infty} e^{-x^2/(2\sigma^2)}\,\mathrm{d}x = \sigma\sqrt{2\pi}.$$

Therefore the Gaussian normalizing constant is $1/(\sigma\sqrt{2\pi})$.

Exercise 7 – Multiple integrals and change of variables. For J_1, direct integration gives

$$J_1 = \int_{-1}^{2}\left(6x - \frac{9}{2}\right)\mathrm{d}x = -\frac{9}{2}.$$

Reversing the order, $0 \le y \le 3$ and $-1 \le x \le 2$, gives the same value. For the triangle D,

$$D = \{0 \le x \le 2,\ 0 \le y \le (2-x)/2\} = \{0 \le y \le 1,\ 0 \le x \le 2-2y\},$$

and either description gives $J_2 = 4/3$. Finally, polar coordinates give

Result

$$J_3(R) = \int_0^{2\pi}\int_1^{R} r^4 r\,\mathrm{d}r\,\mathrm{d}\theta = \frac{\pi}{3}(R^6 - 1), \qquad J_3(3) = \frac{728\pi}{3}.$$

4 Black–Scholes Greeks

Exercise 8 – Derive the Greeks. Recall that

$$\Phi(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-u^2/2}\,\mathrm{d}u, \qquad \phi(x) = \Phi'(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}.$$

With all unmentioned variables held fixed, the Greeks are defined by

$$\Delta = \frac{\partial C}{\partial S}, \qquad \Gamma = \frac{\partial^2 C}{\partial S^2}, \qquad \text{Vega} = \frac{\partial C}{\partial \sigma},$$

$$\Theta = -\frac{\partial C}{\partial T}, \qquad \text{Rho} = \frac{\partial C}{\partial r}.$$

Here T is time to maturity, so the market convention for calendar-time Theta is $-C_T$.

Question 2 – Key identity. Since $d_1 - d_2 = \sigma\sqrt{T}$ and

$$d_1 + d_2 = \frac{2(\log(S/K) + rT)}{\sigma\sqrt{T}},$$

we have

$$\frac{\phi(d_2)}{\phi(d_1)} = \exp\left(\frac{d_1^2 - d_2^2}{2}\right) = \exp\bigl(\log(S/K) + rT\bigr).$$

Therefore

Result

$S\phi(d_1) = Ke^{-rT}\phi(d_2)$.

Question 3 – Closed-form Greeks.

Delta and Gamma. We use

$$\partial_S d_1 = \partial_S d_2 = \frac{1}{S\sigma\sqrt{T}}.$$

The chain rule gives

$$\frac{\partial C}{\partial S} = \Phi(d_1) + \frac{\phi(d_1)}{\sigma\sqrt{T}} - \frac{Ke^{-rT}\phi(d_2)}{S\sigma\sqrt{T}}.$$

The last two terms cancel by the key identity. Differentiating once more,

$$\boxed{\Delta = \Phi(d_1), \qquad \Gamma = \frac{\phi(d_1)}{S\sigma\sqrt{T}}.}$$

Vega. Differentiating both normal-distribution terms with respect to σ and again using the key identity leaves

$$\boxed{\text{Vega} = \frac{\partial C}{\partial \sigma} = S\phi(d_1)\sqrt{T}.}$$

Rho. Since $\partial_r d_1 = \partial_r d_2 = \sqrt{T}/\sigma$, the chain-rule terms cancel and the derivative of the discount factor remains:

$$\boxed{\text{Rho} = \frac{\partial C}{\partial r} = KTe^{-rT}\Phi(d_2).}$$

Theta. Differentiation with respect to time to maturity yields

$$\frac{\partial C}{\partial T} = \frac{S\phi(d_1)\sigma}{2\sqrt{T}} + rKe^{-rT}\Phi(d_2).$$

Consequently, calendar-time Theta is

$$\boxed{\Theta = -\frac{\partial C}{\partial T} = -\frac{S\phi(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}\Phi(d_2).}$$

Question 4 – Short-maturity at-the-money approximation.

(a) At the money forward means $K = Se^{rT}$. Discounting both sides shows that this is equivalent to $Ke^{-rT} = S$.

(b) Substitution in the definitions of d_1 and d_2 gives

$$d_1 = \frac{\sigma\sqrt{T}}{2}, \qquad d_2 = -\frac{\sigma\sqrt{T}}{2}.$$

(c) Since $Ke^{-rT} = S$, the call price becomes

$$C = S\left[\Phi\left(\frac{\sigma\sqrt{T}}{2}\right) - \Phi\left(-\frac{\sigma\sqrt{T}}{2}\right)\right].$$

Using $\Phi(-z) = 1 - \Phi(z)$ and $\Phi(z) = \frac{1}{2} + z/\sqrt{2\pi} + O(z^3)$,

(d) With $z = \sigma\sqrt{T}/2$, the expansion gives

$$C = S\left[2\left(\frac{1}{2} + \frac{\sigma\sqrt{T}}{2\sqrt{2\pi}} + O(T^{3/2})\right) - 1\right].$$

Therefore

Result

$$C_{\text{ATM,fwd}} = S\sigma\sqrt{\frac{T}{2\pi}} + O(T^{3/2}) \text{ as } T \downarrow 0.$$

(e) Interpretation of the $\sqrt{T}$ scale. Over a short horizon T, the typical random move of the underlying in a diffusion model is of size

$$S\sigma\sqrt{T}.$$

An at-the-money option has almost no intrinsic value, so its price is mainly the value of the positive part of this uncertain short-horizon move. This is why its leading-order time value inherits the same $\sqrt{T}$ scale:

$$C_{\text{ATM,fwd}} \sim \frac{1}{\sqrt{2\pi}} S\sigma\sqrt{T}.$$

The dependence on maturity is therefore not linear. Dividing T by four divides the leading-order option value by two; halving T divides it by $\sqrt{2}$, not by two. Equivalently, differentiating the approximation gives

$$\frac{\partial C}{\partial T} \sim \frac{S\sigma}{2\sqrt{2\pi T}}, \qquad \Theta = -\frac{\partial C}{\partial T} \sim -\frac{S\sigma}{2\sqrt{2\pi T}}.$$

Thus the magnitude of time decay becomes very large as $T \downarrow 0$ at the money. This singular behavior is specific to the immediate neighborhood of the strike; away from the money, the short-maturity asymptotics are different.

5 Numerical Schemes for ODEs

Exercise 9 – Euler accuracy and the classical inequalities. For $y' = -2y$,

$$y_{n+1}^E = (1-2h)y_n^E, \qquad y_{n+1}^I = \frac{1}{1+2h} y_n^I,$$

so

Result

$$y_n^E = (1-2h)^n, \qquad y_n^I = (1+2h)^{-n}, \qquad y(t_n) = e^{-2nh}.$$

Both methods have global error $O(h)$, so halving h asymptotically halves the maximum error. Explicit Euler is stable precisely when $|1-2h| < 1$, hence $0 < h < 1$. The implicit multiplier satisfies $0 < (1+2h)^{-1} < 1$ for every $h > 0$. Stability controls growth, not the size of the discretization error; a large stable step can therefore be inaccurate.

6 Solving Scalar and Multidimensional ODEs

Exercise 10 – Scalar linear equations. Applying variation of constants or the integrating-factor method gives

Result

$$y_1(t) = \frac{t}{2} - \frac{1}{8} - \frac{7}{8}e^{-4t}, \qquad y_2(t) = (1+t)\left(2 + t + \frac{t^2}{2}\right), \qquad y_3(t) = e^{2t} - e^t.$$

For the second equation, the integrating factor is $(1+t)^{-1}$ and $[y/(1+t)]' = 1+t$. Direct differentiation verifies all three initial values and equations (the second solution is considered on the interval $t > -1$).

Exercise 11 – Nonlinear scalar equations. Separation of variables gives

Result

$$y' = y(1-y): \quad y(t) = \frac{1}{1 + \frac{1-y_0}{y_0} e^{-t}}, \qquad t \in \mathbb{R}.$$

The equilibria are 0 (unstable) and 1 (asymptotically stable). Similarly,

$$y' = -y^3: \quad y(t) = \frac{1}{\sqrt{1+2t}}, \qquad t \in \left(-\frac{1}{2}, \infty\right).$$

Its only equilibrium is 0, which is asymptotically stable, although the linearization vanishes. Finally,

$$y' = t(1+y^2): \quad \arctan y = \frac{t^2}{2}, \qquad y(t) = \tan\left(\frac{t^2}{2}\right).$$

The maximal interval containing zero is $(-\sqrt{\pi}, \sqrt{\pi})$. The last equation is nonautonomous, so a one-dimensional autonomous phase line is not applicable to it.

Exercise 12 – Linear systems in dimension two. The eigenpairs may be chosen as

$$\lambda_1 = -2,\ v_1 = (1,0)^\top, \qquad \lambda_2 = -4,\ v_2 = (1,-2)^\top.$$

Since $x(0) = 2v_1 - v_2$,

Result

$$x(t) = 2e^{-2t}v_1 - e^{-4t}v_2 = \begin{pmatrix} 2e^{-2t} - e^{-4t} \\ 2e^{-4t} \end{pmatrix}.$$

This is exactly $e^{tA}x(0)$. The numerical updates are

$$x_{n+1}^E = (I + hA)x_n^E, \qquad x_{n+1}^I = (I - hA)^{-1}x_n^I.$$

Their errors against the displayed exact solution decay linearly with h.

Exercise 13 – Forced multidimensional system. Solving $Ax^\star + b = 0$ gives $x^\star = (1,1)^\top$. With $z = x - x^\star$, $z' = Az$ and $z(0) = (-1,1)^\top$. Componentwise, $z_1 = -e^{-2t}$ and $z_2 = e^{-2t}$, hence

Result

$$x(t) = \begin{pmatrix} 1 - e^{-2t} \\ 1 + e^{-2t} \end{pmatrix} \longrightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix} = x^\star.$$

Explicit Euler uses $x_{n+1} = (I + hA)x_n + hb$; implicit Euler uses $x_{n+1} = (I - hA)^{-1}(x_n + hb)$. Both converge to the same equilibrium when their stability conditions are respected.

Exercise 14 – Second-order equation as a system. The characteristic polynomial is $(r+1)(r+3)$, so the initial data yield

Result

$y(t) = e^{-t} - e^{-3t}$.

With $z = (y, y')^\top$,

$$z' = \begin{pmatrix} 0 & 1 \\ -3 & -4 \end{pmatrix} z, \qquad z(0) = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

The companion matrix has the same eigenvalues -1 and -3. Explicit Euler is $z_{n+1} = (I + hA_c)z_n$; comparing its first component with $e^{-t_n} - e^{-3t_n}$ verifies first-order convergence.